

HENRY (YUAN-HEN) CHIU

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Portfolio: <https://henrychiu0826.github.io/>

PROFILE

Mechatronics master's student specializing in safety-critical PCB design and embedded systems. Three years in FSAE electronics, including leading the team through Melbourne University Racing's first 100% completion of all dynamic events. Experienced across the full hardware cycle: schematic to layout, procurement to integration, bench test to competition.

RELEVANT EXPERIENCE

Melbourne University Racing (MUR) | FSAE Electric — Low Voltage Electronics

Jan 2024 – Present

PCB Designer

2024

- Designed and manufactured the **Master Shutdown System**, the safety-critical PCB that processes emergency shutdown signals (IMD, BMS, BSPD, PDOC) and isolates the accumulator on fault, achieving **100% completion of scrutineering** and the car's first competitive lap in seven years
- Diagnosed an intermittent component failure during vehicle integration under competition deadlines; repaired and integrated a fallback board, then fed lessons into a full redesign

Electronics Team Lead

Dec 2024 – Dec 2025

- Led a team of **5 student engineers** covering all sensors, wiring, and PCBs on the vehicle. The team passed all inspections on schedule and achieved **100% completion of all dynamic events in 2025**
- Collaborated with 10+ cross-disciplinary members to design and integrate the full vehicle wiring harness, optimizing packaging and accessibility for installation and maintenance
- Developed a platform documenting electrical connections across Excel, LucidsSpark, and RapidHarness, improving accessibility and team collaboration
- Standardized inventory and part tracking through PartsBox, preventing duplicate purchases (\$300+ saved); coordinated 30+ meetings end-to-end

Senior PCB Designer

2026

- Designed **STM32-based sensor nodes** for accumulator temperature monitoring and a redesigned, more compact shutdown system; 3+ unique PCBs completed in Altium Designer
- Trained 4 new members in PCB design fundamentals, providing design reviews and advice

Mechatronics Engineering Intern | Clutch Industries

Jul 2025 – Oct 2025

- Designed a **modular water dynamometer** in Fusion 360 to test 4+ motors, 3 propellers, and 2 jet pumps, enabling rapid reconfiguration for comparative thrust testing
- Specified components for sustained underwater loading; researched suppliers and sourced cost-effective parts (up to \$500 saved), managing procurement with JLCMC and Alibaba vendors and coordinating CNC machining to specification
- Delivered a complete, manufacturable design with parts ordered; the rig was subsequently assembled and commissioned by the team and used for live testing

Secretary | Card Crew, University of Melbourne

Sep 2023 – Aug 2024

- Organised weekly events for 20+ attendees and managed club communications and logistics

EDUCATION

Master of Mechatronics Engineering | University of Melbourne

Feb 2025 – Present

WAM: 85.9

Bachelor of Science — Mechatronic Systems | University of Melbourne

Feb 2022 – Nov 2024

WAM: 87.5

Awards: Dean's Honour List 2022 & 2023 · IEEE Award 2023 · Melbourne International Undergraduate Scholarship (full scholarship)

SKILLS

PCB Design: Altium Designer, KiCad — schematic capture, layout, bench bring-up and debug

Embedded: STM32, sensor interfacing, signal conditioning, C

CAD: Fusion 360, SolidWorks — assemblies, DFM, CNC drawings

Programming: Python, C, C++, Java, MATLAB, Excel VBA

Leadership: Team management (5 engineers), cross-team coordination, procurement, documentation

Languages: English, Mandarin